Discrete-Price Strict-h=0 Equilibrium at $(\tau, \gamma) = (0.1, 0.01)$ Sweep, Contours, and Regression

REZN project, $K = 3$, CRRA, no noise traders

1 Setup

Same model and solver as the τ=2.0 companion note (which see for the math). At $\tau = 0.1$ the per-signal precision is 20× weaker: each $u_k \sim \mathcal{N}(v-\frac{1}{2}, 10)$. The FR sigmoid $P_{\mathrm{FR}}(u) = \sigma(\tau \sum_k u_k) = \sigma(0.1 \sum_k u_k)$ is nearly affine over the active signal range, so the model is very close to no-revelation.

We test whether the discrete-price strict-h=0 partition equilibrium still exists in this weak-signal regime, and how it relates to the kernel-h>0 limit and to FR.

2 Sweep results

Table 1: Tail-mean deficit $1 - R^2$ for the log-odds–vs–$\sum u_k$ regression at $\tau = 0.1$. Rightmost column: kernel-h>0 deficit at the same G. "Conv" = exact convergence; "Per-2" = stable period-2 cycle.

$G \backslash M$	8	16	32	64	kernel-h
8	0.423 (Conv)	0.238 (Conv)	0.149 (Conv)	0.074 (Per-2)	0.196
11	0.382 (Conv)	0.230 (Conv)	0.167 (Conv)	0.073 (Conv)	0.196
15	0.395 (Conv)	0.231 (Conv)	0.167 (Conv)	0.061 (Per-2)	0.196
21	0.406 (Conv)	0.236 (Conv)	0.161 (Conv)	0.049 (Conv)	0.196

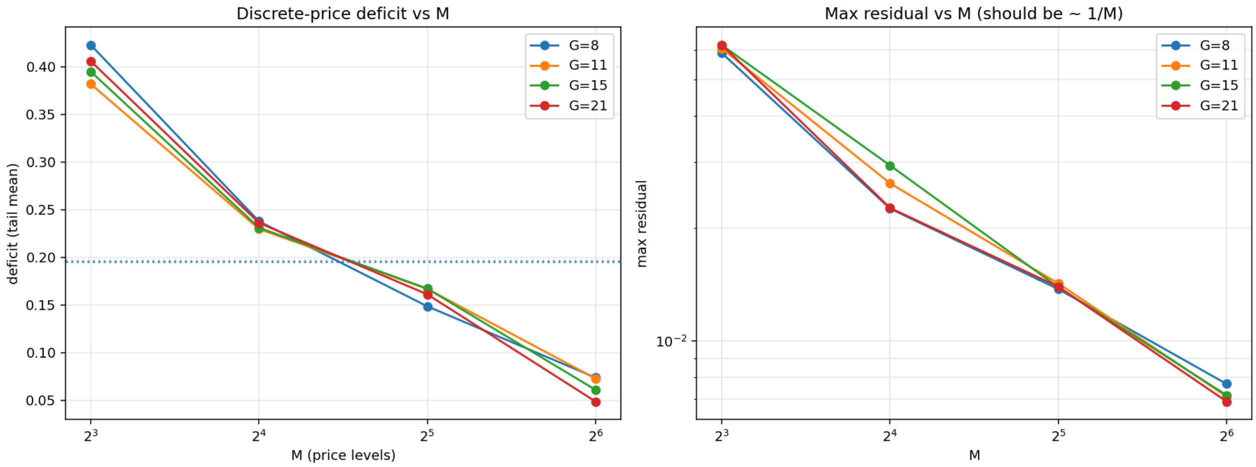

Figure 1: $\tau = 0.1$: left, deficit (tail-mean) vs M per G; dotted horizontals = kernel-h>0 deficits. Right: max residual, floored at $\frac{1}{2}\Delta p$.

Reading the table. At $\tau = 0.1$ the deficit is *monotone decreasing in* M, from ~ 0.40 at $M = 8$ down to ~ 0.05 at $M = 64$ — the discrete-price field is converging toward an essentially affine (near-FR) log-odds relation as the price grid refines. This is qualitatively different from $\tau = 2.0$, where the deficit sat in $[0.17, 0.25]$ across M with no clear monotone behaviour.

Comparison with $\tau = 2.0$. Both regimes show:

- Strict-h=0 partition equilibria exist for every (G, M);
- Max residual hits the discretization floor $\frac{1}{2}\Delta p$;
- At small M the discrete deficit exceeds the kernel deficit.

They differ in their $M \to \infty$ asymptote: at $\tau = 2.0$ the limit appears to plateau below the kernel (~ 0.18 vs 0.29); at $\tau = 0.1$ the limit appears to head toward zero (the FR plane). This is consistent with weak signals — the partition has little information to recover, so refining the grid quickly traces the FR relation.

3 Contour slices – $G = 21$

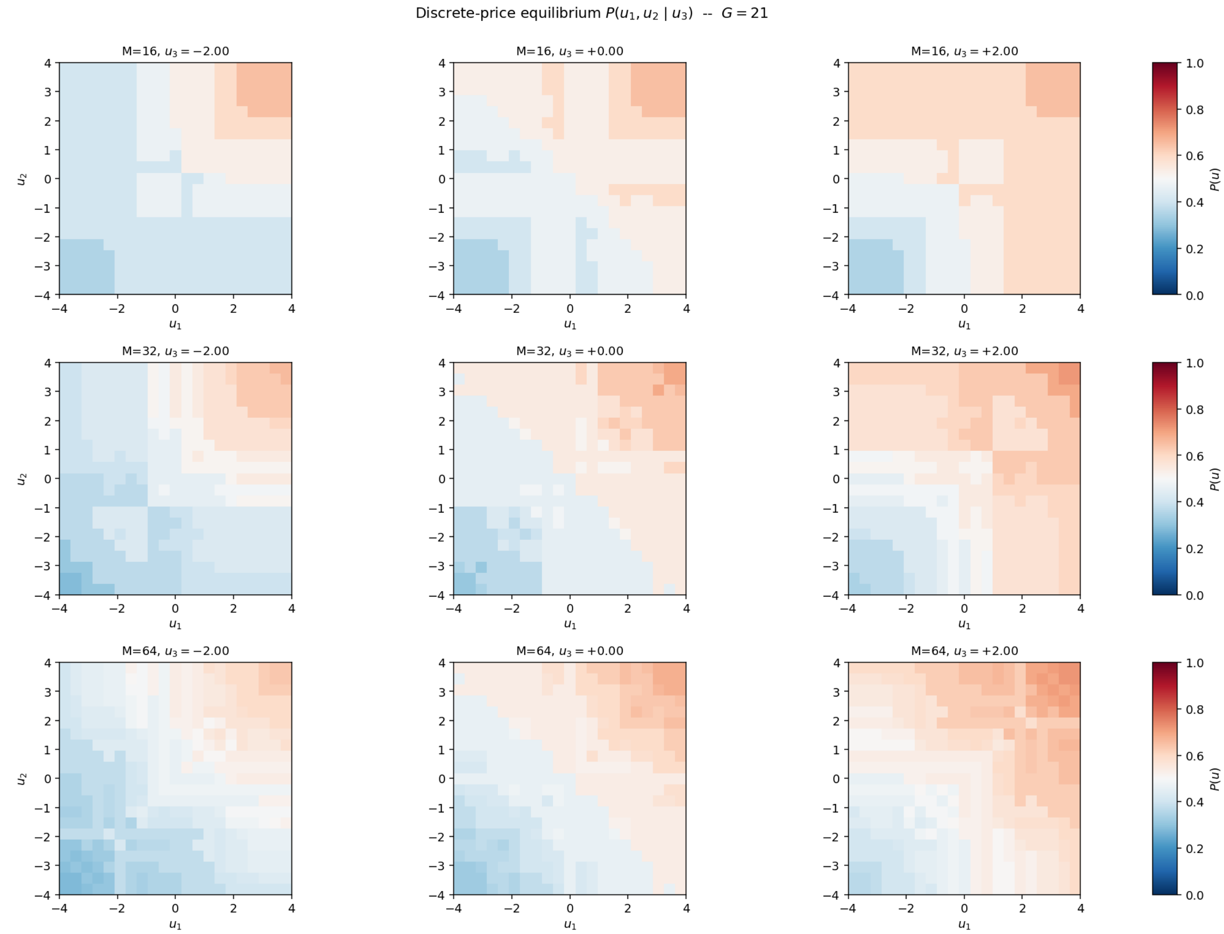

Figure 2: Discrete-price $P(u_1, u_2 \mid u_3)$ at $G = 21$, $\tau = 0.1$. Rows: $M = 16, 32, 64$. Columns: $u_3 \approx -2, 0, +2$. At low τ the field is much smoother across u: most cells sit near $P = 0.5$ because signals carry little information about v.

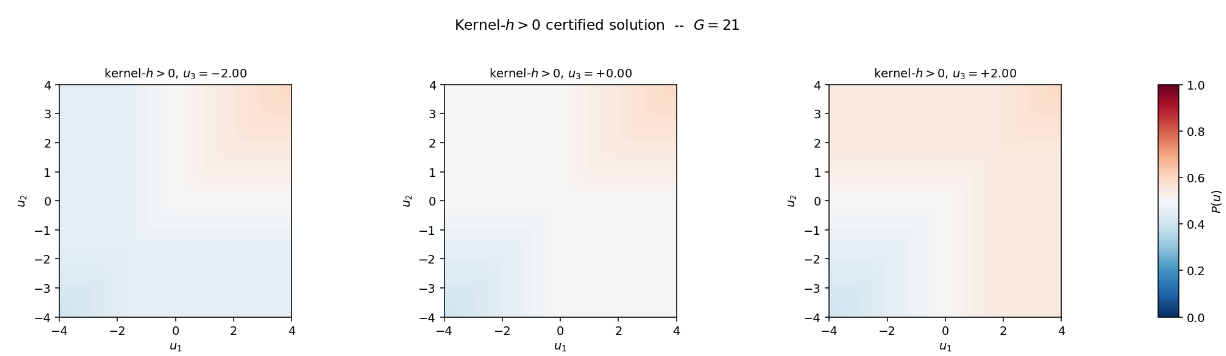

Figure 3: Kernel-h>0 certified solution at $G = 21$, $\tau = 0.1$ for comparison. The kernel field is again smooth; the discrete-price fields quantise it.

4 Log-odds regression – $G = 21$

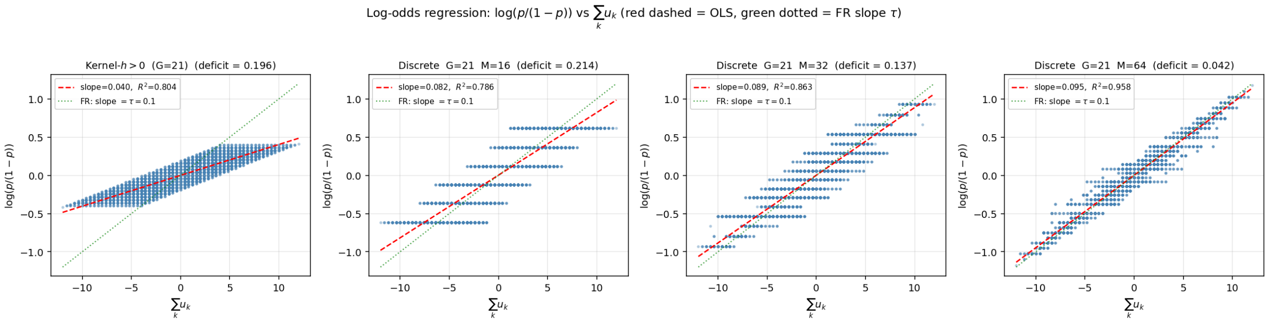

Figure 4: Log-odds vs $\sum u_k$ at $G = 21$, $\tau = 0.1$. Leftmost: kernel-h>0; remaining three: discrete-price with $M = 16, 32, 64$. Red dashed: OLS fit. Green dotted: FR slope $\tau = 0.1$. At this τ the kernel OLS slope is much closer to the FR line than at $\tau = 2.0$, and the discrete-price OLS slope tracks FR closely at large M.

5 Takeaways

- **Discrete-price equilibria still exist at low τ.** Every (G, M) cell finds either an exact converged or stable period-2 fixed point; the residual hits the discretization floor.
- **Different asymptotic behaviour in M.** At $\tau = 0.1$ the deficit decreases monotonically with M, indicating the limit object is close to the FR plane. At $\tau = 2.0$ the deficit plateaus, indicating a genuinely non-FR limit.
- **Cleaner convergence at low τ.** No period-k instability beyond $M = 64$; K-means converges in a few iterations at every (G, M).
- **Strict-h=0 formulation is robust.** The discrete-price construction yields a valid noise-free strict-h=0 equilibrium across both regimes, confirming the result is structural, not an artifact of one parameter choice.

Data & code.

- Sweep: `TAU=0.1 OUT=...lowtau/discrete_price_sweep_tau01 python3 discrete_price_sweep.py`
- Viz: same env, `make_viz_integrated.py`
- Results: `projects/REZN/solved_fixed_points/lowtau/discrete_price_sweep_tau01/`